Roll No.: .....

END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B.Tech(CSE) | Semester: IV

Name of the Course: Fundamentals of Stastics and AI | Course Code: TCS421

Time: 3 Hours | **Maximum Marks: 100**

Note:

i. **All questions are compulsory.**

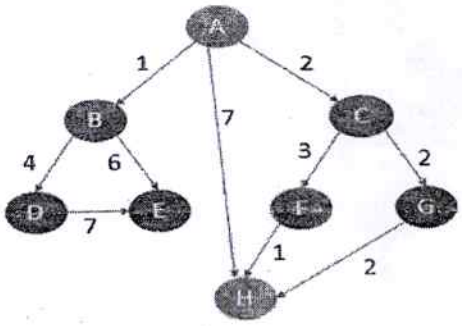
ii. Answer any two sub questions among a, b & c in each main questions

iii. Total marks in each main question are twenty.

iv. Each questions carries 20 marks

Q1	(20Marks)	CO1
(a)	What is the goal of AI? Explain the advantages and disadvantages of AI.	CO1
(b)	Describe the concept of Intelligent Systems and application of AI in detail. Explain the concept of PEAS Representation.	CO2
(c)	Define informed Search? Discuss Best First Search in detail with suitable example.	CO2
Q2		CO2
(a)	What is searching? Explain different types of searching algorithm used in AI.	CO2
(b)	Discuss A* search algorithm and apply the same for the given graph to search the goal node. Consider Start Node=S and Goal Node=G.	CO2

NODES	(Start Node) A	B	C	D	E	F	G	(Goal Node) H
H(N)	5	3	4	2	6	3	1	0



(c)	What are the five components of problem formulation for a problem-solving agent? How problems can be solved by problem-solving agents?	
Q3		
(a)	What is the fundamental difference between data science and business intelligence (BI)?	CO4
(b)	How do you perform basic exploratory data analysis (EDA) in Pandas?	CO3
	Suppose we have the following data:	CO3

Hours Studied	Exam Score
2	50
4	60
6	70
8	75

	10 | 85 Create a scatter plot in python.	
(c)	Describe the roles and responsibilities of a data scientist. Discuss the key functions and duties performed by a data scientist.	
Q4		CO5
(a)	How do you handle missing data in R? Explain barplot, linechart using R programming with the help of an example.	CO5
(b)	Difference between Data Pre-processing and Data Cleaning. What are the steps involved in preprocessing data? Why is preprocessing important in data?	CO4
(c)	What is the difference between underfitting and overfitting a model? How do you evaluate whether a model fits the data well?	
Q5		
(a)	What are the basic descriptive statistics used to summarize a dataset. How do variance and standard deviation measure data spread?	CO5
(b)	Difference between Linear Regression and Logistic regression. In what types of problems would you use logistic regression instead of linear regression?	CO6
(c)	What is the Pearson Correlation Coefficient? Compute the Pearson Correlation Coefficient from the following table:	CO6

Age (X)	Glucose Level (Y)
43	99
21	65
25	79
42	75
57	87
59	81